



Towards Controllable Hybrid Fairness in Graph Neural Networks

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Abstract

Graph Neural Networks (GNNs) have shown remarkable capabilities in mining graph-structured data. However, conventional GNNs often encounter various fairness issues, such as predictions with prejudices when dealing with nodes with different sensitive attributes like genders or races, or significantly different prediction performance when facing nodes with different degrees. Existing studies mainly focus on addressing one specific fairness issue, neglecting the fact that a GNN model may face multiple unfairness simultaneously in reality, and addressing only one specific fairness may still leave the GNNs in an unfair status.

In this paper, we focus on achieving multiple fairness on GNNs simultaneously, which we call hybrid fairness. To achieve this objective, we propose a novel GNN framework called **LibraGNN**. Specifically, we adopt a multi-teacher knowledge distillation training framework that successfully unifies the learning paradigms for multiple fairness. To ensure LibraGNN strikes a better trade-off among different fairness, we transform the multi-teacher knowledge distillation into a multi-objective optimization problem and further employ Pareto efficiency for optimization guidance. Finally, a controllable preference vector is introduced to assist LibraGNN in modulating its capability towards various forms of fairness, thereby achieving controllable hybrid fairness. Extensive experiments on three real-world datasets demonstrate the effectiveness of LibraGNN on both hybrid fairness and utility.

CCS Concepts

• Computing methodologies → Artificial intelligence.

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Keywords

Graph Neural Networks; Hybrid Fairness; Pareto Optimal

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1 Introduction

With the strong capability in mining graph-structured data, *Graph Neural Networks* (GNNs) have demonstrated remarkable expressive power in many domains such as social network analysis [32] and bioinformatics [27]. However, despite their notable achievements, GNNs frequently encounter a myriad of fairness issues, i.e. the outcomes of the GNNs are discriminative against certain groups. For example, in hiring web platforms, the predictions of GNNs are highly related to sensitive node attributes like race and gender [7, 24], while in recommendations scenarios, GNNs tend to provide more accurate prediction services for highly active users compared to inactive users [31].

To address fairness issues in GNNs, prior research has proposed various strategies, such as incorporating adversarial learning modules [7], data augmentation [20, 23], and other methods. However, these methods mainly focus on limited fairness scenarios, disregarding the possibility of multiple forms of unfairness coexisting. Consequently, although these methods may effectively tackle specific fairness concerns in GNNs, from the perspective of other fairness dimensions, the GNNs may still remain unfair. For example, in the context of e-commerce, despite the GNN-based recommendation system striving for equal treatment of different gender groups, an inherent disparity in the GNN-based recommender persists when dealing with highly active users as opposed to low-activity users, exemplified by challenges such as the cold start problem.

Conceptually, we refer to the status of achieving multiple kinds of fairness as **hybrid fairness**, which actually brings several non-trivial challenges. 1) *How to address various types of unfairness with a unified approach?* Different types of unfairness may have different underlying causes [22], requiring distinct strategies for effective mitigation. For example, the degree-related unfairness actually stems from neighborhood structure, making debiasing constraints

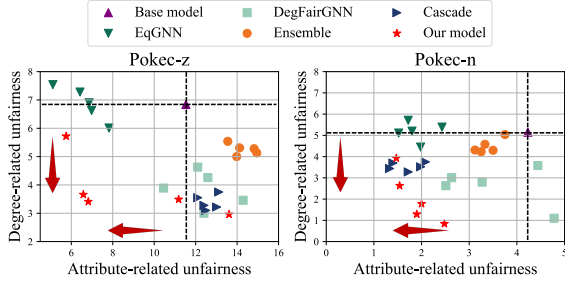


Figure 1: Performance of existing models on different fairness. DEO is the fairness metric and lower values are better.

in achieving attribute-based fairness infeasible [22]. Moreover, simply combining these strategies in one model may lead to an overly complex model architecture and restrict its expandability to other forms of unfairness. Intuitively, simple combinations of existing fair GNNs such as Ensemble or Cascade may be reasonable solutions for the above limitation, but they actually face a new challenge: 2) *How to balance the trade-off among multiple fairness?* Different forms of fairness are not mutually independent. In this context, it is hard to achieve the best results on all fairness at the same time, and the concept of hybrid fairness essentially means achieving a trade-off among multiple fairness. Unfortunately, the majority of existing approaches fail to realize this issue, resulting in their monotonous performance across various fairness. Figure 1 further supports this claim with empirical evidence. With considering two kinds of fairness, i.e. attribute-related fairness and degree-related fairness, Figure 1 illustrates the fairness performance of GCN [16] (base model), DegFairGNN [22], EqGNN [29], and their heuristic combination through Ensemble and Cascade approaches. It can be seen that all existing methods achieve trade-offs between two fairness only within a limited scope. For example, all results of EqGNN significantly reduce attribute-related unfairness on the Pokcc-z dataset, but the impact on degree-related unfairness compared to the base model remains limited. Such trade-off in a limited scope will not only fall into the predicament of optimizing single fairness but also fail to fulfill the diverse fairness requirements in reality.

To address these issues, we propose a novel GNN framework namely **LibraGNN**. Specifically, we first adopt a multi-teacher knowledge distillation framework to unify the solutions for various forms of unfairness, allowing the GNN model to draw relevant knowledge from teachers specialized in different fairness aspects. Next, to formulate the trade-offs among multiple fairness, we transform the multi-teacher knowledge distillation into a multi-objective optimization problem and introduce Pareto efficiency to guide the optimization. Finally, to help the student model find more diverse pareto solutions, a preference vector module is proposed to effectively control the student GNN model’s preference towards different teachers. Extensive experiments on three real-world datasets consistently demonstrate that LibraGNN can significantly improve hybrid fairness while keeping the high accuracy of models. Our contributions can be summarized as follows:

- We propose a multi-teacher knowledge distillation framework LibraGNN to achieve hybrid fairness in a unified manner. Instead of an overly complex model to handle different unfairness, LibraGNN is flexible and easy to deploy.

- We first utilize Pareto efficiency to assist LibraGNN in balancing the trade-offs among different fairness. A controllable preference module is further proposed to adjust LibraGNN’s capability towards different fairness, thus achieving diverse solutions.
- Comprehensive experiments are carried out on three real-world datasets, which verify the effectiveness and expandability of LibraGNN in achieving hybrid fairness.

2 Related Work

Fairness in GNNs. Extensive research has verified the unfairness issues in GNN models, which would inevitably lead to some social risks [7, 23, 34] or sub-optimal performance [21, 25]. Specifically, FairGNN [7], CFC [2], and UGE [34] utilize adversarial learning to obtain node embeddings that are independent of sensitive attributes. Wang et al. [35], Dong et al. [10], and Yang et al. [37] further consider the effect of message propagation on causing unfairness in attributed networks. Ling et al. [20] and Luo et al. [23] try to improve GNN fairness through data augmentation. Similar to our work, Dong et al. [11] and Zhu et al. [43] also utilize knowledge distillation for mitigating fairness issues, but they only focus on addressing one specific kind of fairness. Besides sensitive attributes, many researchers also try to improve GNNs’ fairness related to the graph structure, such as mitigating the unfairness between tail nodes and head nodes [21, 22] or among different node clusters [25]. Recently, a branch of researchers began to focus on the fairness issues from multiple dimensions [4, 6, 39]. However, their scopes are still limited to specific scenarios and lack expandability to other forms of fairness like individual fairness [9, 30] or counterfactual fairness [1].

Multi-teacher Knowledge Distillation. Knowledge distillation is first proposed for model compression [14], which aims to distill the knowledge in an expressive yet cumbersome model (teacher) to a lightweight model (student). Recently, due to the ability to transfer diverse knowledge to student models simultaneously, multi-teacher knowledge distillation also attracts lots of focus [8, 42]. For example, to help the student have good performance in both normal and biased environments, Ding et al. [8] use two teacher models to distill corresponding knowledge in an interpolative manner. Similarly, by using different teacher models, Zhao et al. [42] successfully improve the robustness of the student model while maintaining the accuracy on normal samples in an adversarial-distillation manner. In this work, we aim to learn how to achieve different kinds of fairness from multiple fair teacher GNN models, thus achieving hybrid fairness.

3 Preliminary

3.1 Problem Statement

Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, it would contain $|\mathcal{V}|$ nodes and $|\mathcal{E}|$ edges. The feature matrix for the graph is denoted as $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times D}$, where D represents the feature dimension of each node. Let $\mathcal{Y}$ denote the set of node labels, and each node has one unique label $y \in \mathcal{Y}$. Formally, concentrating on group fairness², given K different kinds of fairness, for the k th kind of fairness, the nodes

²Group fairness refers to the concept of treating different groups of individuals or entities (like "male" and "female", "head nodes" and "tail nodes") equally.

set $\mathcal{V}$ can be split into C^k disjoint groups $\{\mathcal{V}_1, \mathcal{V}_2, \dots, \mathcal{V}_{C^k}\}$ where $\bigcup_{i=1}^{C^k} \mathcal{V}_i = \mathcal{V}$, and $\mathcal{V}_i \cap \mathcal{V}_j = \emptyset \ \forall i, j \in \{1, 2, \dots, C^k\}$ and $i \neq j$. Note that, the expandability to other kinds of fairness like counterfactual fairness [1] will also be discussed in the experiments.

Our goal is to design a model-agnostic framework that can enhance the hybrid fairness of GNNs. In detail, given a base GNN $\mathcal{A}$, we can obtain an enhanced version of the GNN $\hat{\mathcal{A}}$ that exhibits good fairness on K different kinds of scenarios simultaneously, i.e. maximizing the *hybrid fairness index* (HFI), while the utility of the GNN model is preserved. The definition of HFI will be given later.

3.2 Fairness Evaluation Metrics

In this section, we will present the evaluation metrics for fairness in detail. We begin by focusing on the scenario where only one kind of fairness is considered. Following the line of [13, 22], we list the *Difference of Statistical Parity* (DSP), *Difference of Equal Opportunity* (DEO), and *Difference of Counterfactual Parity* (DCP) to quantitatively assess one specific kind of fairness.

DEFINITION 1. *Difference of the Statistical Parity (DSP)*: Given the k th kind of fairness, $\forall i, j \in \{1, 2, \dots, C^k\}$ and $i \neq j$, the predictions should be independent of the group membership, and the corresponding DSP_k can be formulated as:

$$\text{DSP}_k = \max_{i,j} |\mathbb{E}[P(\hat{y}_v = y|v \in \mathcal{V}_i)] - \mathbb{E}[P(\hat{y}_v = y|v \in \mathcal{V}_j)]|. \quad (1)$$

DEFINITION 2. *Difference of the Equal Opportunity (DEO)*: Given the k th kind of fairness, $\forall i, j \in \{1, 2, \dots, C^k\}$ and $i \neq j$, the probability of an instance being predicted into its true class should be independent of the group membership, and the corresponding DEO_k can be formulated as:

$$\text{DEO}_k = \max_{i,j} |\mathbb{E}[P(\hat{y}_v = y|y_v = y, v \in \mathcal{V}_i)] - \mathbb{E}[P(\hat{y}_v = y|y_v = y, v \in \mathcal{V}_j)]|. \quad (2)$$

$\hat{y}_v$ denotes the predicted class of node v . Both DSP and DEO are used for evaluating group fairness. Specifically, DSP asks the prediction probability to be independent of the group membership of nodes, while DEO actually represents the maximum accuracy difference among groups. In addition, we employ DCP for assessing counterfactual fairness.

DEFINITION 3. *Difference of the Counterfactual Parity (DCP)*: Given the k th kind of fairness, the protected attribute of node v is denoted as s_v , and other attributes are denoted as x_v , then the corresponding DCP_k can be formulated as:

$$\text{DCP}_k = \max_{s_v, s_v'} |\mathbb{E}[P(\hat{y}_v = y|s_v, x_v)] - \mathbb{E}[P(\hat{y}_v = y|s_v', x_v)]|. \quad (3)$$

s_v' is the counterfactual protected attribute of node v . Since we aim to improve multiple fairness of a GNN model, with borrowing the concept of average relative improvement in multi-task learning [33], we further define *Hybrid Fairness Index* HFI($\mathcal{A}, \hat{\mathcal{A}}$) to evaluate the hybrid fairness of an enhanced GNN model $\hat{\mathcal{A}}$ compared with its base model $\mathcal{A}$.

DEFINITION 4. *Hybrid Fairness Index (HFI)*: Considering the settings of K kinds of fairness, if there is a base model $\mathcal{A}$ with fairness metric F_k and an enhanced GNN model $\hat{\mathcal{A}}$ with $\hat{F}_k$ on the k th kind

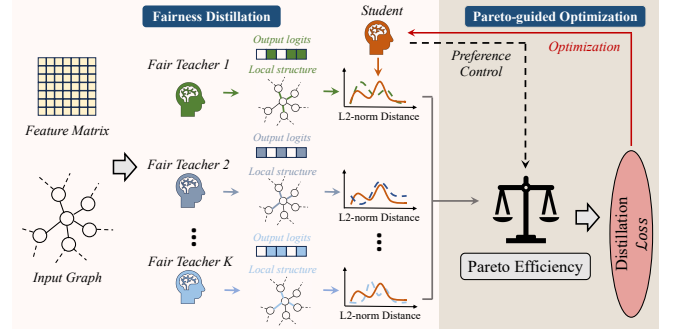


Figure 2: The overall framework of LibraGNN

of fairness, respectively, then the Hybrid Fairness Index (HFI) of GNN model $\hat{\mathcal{A}}$ can be formulated as:

$$\text{HFI}(\mathcal{A}, \hat{\mathcal{A}}) = \frac{1}{K} \sum_{k=1}^K (F_k - \hat{F}_k) / F_k. \quad (4)$$

Note that, the definition of F_k is flexible, which could be DSP, DEO, DCP, and other fairness metrics. The specific definition of HFI can be flexibly configured by the practitioners based on their real-world requirements.

3.3 Pareto-related Concepts

For a better understanding of our work, we introduce some basic concepts on Pareto efficiency here. As an important concept in multi-objective optimization, Pareto efficiency means a situation in which it is impossible to improve one objective without hurting other objectives. Specifically, we have the following definitions:

DEFINITION 5. *Pareto Dominance*: Given a model which aims at optimizing a series of objective functions $\mathcal{L}_1, \mathcal{L}_2, \dots, \mathcal{L}_K$ (for minimization), assume θ^a and θ^b are two solutions of this model. Then θ^a dominates θ^b ($\theta^a < \theta^b$) if and only if $\mathcal{L}_i(\theta^a) \leq \mathcal{L}_i(\theta^b)$, $\forall i \in \{1, 2, \dots, K\}$, and $\mathcal{L}_j(\theta^a) < \mathcal{L}_j(\theta^b)$, $\exists j \in \{1, 2, \dots, K\}$.

Based on the definition of Pareto dominance, we can have:

DEFINITION 6. *Pareto Efficiency*: θ^a is said to be Pareto efficient if and only if there is no solution θ^b that can dominate θ^a . All Pareto efficient points consist of a Pareto optimal set.

4 Methodology

In this section, we give the details of LibraGNN. The overall framework of LibraGNN is illustrated in Figure 2. To address multiple kinds of fairness issues in GNNs in a unified manner, we first adopt a multi-teacher knowledge distillation approach to extract fairness-related knowledge from various existing fair models. After that, to better balance the trade-offs among different kinds of fairness, we turn the multi-teacher knowledge distillation into a multi-objective optimization problem and utilize Pareto efficiency for optimization guidance. At last, a controllable mechanism is introduced to regulate the student's preference toward different teachers, thus achieving diverse hybrid fairness. In the following, we will elaborate on each component of LibraGNN.

4.1 Hybrid Fairness through Distillation

Fairness Distillation. With the presence of various models addressing different unfairness, one key assumption of our work is that these existing fair models have learned corresponding fairness-related knowledge. In this way, through utilizing a multi-teacher knowledge distillation mechanism to concurrently extract versatile fairness-related knowledge from these existing fair models, we can naturally achieve hybrid fairness in a unified manner. In this setting, these existing fair models are called teachers, while the base GNN that will learn fairness-related knowledge from teachers is known as the student. To formulate our distillation, similar to [26], we first use the L2 norm to match the output logits between the student and teachers. Other distillation loss functions like KL-divergence and MMD loss will also be discussed in the experiments and we choose L2-norm loss due to its demonstrated good performance and its facilitation of proofs for key lemmas. Specifically, let $\mathbf{P}_{i,k}^T \in \mathbb{R}^{1 \times |\mathcal{Y}|}$ denote one sample's post-softmax output logits of the teacher model that specializes in the k th kind of fairness. We define the first knowledge distillation loss function as:

$$\mathcal{L}_k^1 = \frac{1}{N} \sum_i \|\mathbf{P}_i^S - \mathbf{P}_{i,k}^T\|_2, \quad (5)$$

where N is the number of samples, $\mathbf{P}_i^S \in \mathbb{R}^{1 \times |\mathcal{Y}|}$ represents the post-softmax output logits of student GNN model, and $\|\cdot\|_2$ denotes the L2 norm distance. Intuitively, as the logits of the student model and teacher model converge, the merits of addressing a specific kind of unfairness that the teacher model excels at are successfully inherited by the student model. Additionally, prior work [18] indicates that message propagation also plays an important role in GNN fairness, and to fully utilize the useful fair knowledge from teacher models, we introduce the second knowledge distillation loss to preserve the latent local structure of teacher models [38]:

$$\mathcal{L}_k^2 = \frac{1}{N} \sum_i \left\| \left(\text{softmax}_{(i,j) \in \mathcal{E}}(\mathcal{K}(\mathbf{P}_i^S, \mathbf{P}_j^S)) - \text{softmax}_{(i,j) \in \mathcal{E}}(\mathcal{K}(\mathbf{P}_i^T, \mathbf{P}_j^T)) \right) \right\|_2 \quad (6)$$

where $\mathcal{K}(\cdot, \cdot)$ is the kernel function, and we use Euclidean distance here. Eq. (6) preserves the message propagation pattern of each node $i \in \mathcal{V}$ to enhance the effectiveness of knowledge distillation. To sum up, the overall knowledge distillation loss function is:

$$\mathcal{L}_k = \mathcal{L}_k^1 + \alpha * \mathcal{L}_k^2, \quad (7)$$

where α is a weight hyper-parameter. Since our goal is to train a student GNN model that can achieve different kinds of fairness without compromising its utility, the objective function is formulated as:

$$\begin{aligned} \min_{\theta} \quad & w_{ce} \mathcal{L}_{CE}(\theta) + \sum_{k=1}^K w_k \mathcal{L}_k(\theta) \\ \text{s.t.} \quad & w_{ce} + \sum_{k=1}^K w_k = 1, w_{ce} > 0, w_k > 0, \forall k \in \{1, \dots, K\}, \end{aligned} \quad (8)$$

where θ is model parameters of the student GNN model, and $\mathcal{L}_{CE}(\theta)$ denotes the normal cross entropy loss. K represents the number of fairness scenarios, and w_{ce} , and w_k are normalized tunable weights to control the impact of each term.

Objective Simplification. In fact, lots of prior works in pursuing fairness of GNNs can successfully preserve a competitive accuracy [7, 22], which indicates that they may naturally contain

useful knowledge on achieving high utility. To verify if there is any knowledge redundancy during knowledge distillation, and further simplify our objective functions, we theoretically analyze the relationship between utility-oriented objective, $\mathcal{L}_{CE}$ and our first knowledge distillation objective $\mathcal{L}_k^1$.

LEMMA 1. *Given a teacher model with high utility, whose output logits are $\mathbf{P}_k^T \in \mathbb{R}^{N \times |\mathcal{Y}|}$, then minimizing objective $\mathcal{L}_k^1$ will lead to the minimization of $\mathcal{L}_{CE}$ between student model output $\mathbf{P}^S \in \mathbb{R}^{N \times |\mathcal{Y}|}$ and the one-hot true labels $\mathbf{P}^* \in \mathbb{R}^{N \times |\mathcal{Y}|}$.*

The proof is given in the Appendix A.2. In this light, our knowledge distillation objectives could also ensure the utility of the student model and we could simplify Eq. (8) to:

$$\begin{aligned} \min_{\theta} \quad & \sum_{k=1}^K w_k \mathcal{L}_k(\theta) \\ \text{s.t.} \quad & \sum_{k=1}^K w_k = 1, w_k > 0 \quad \forall k \in \{1, 2, \dots, K\}. \end{aligned} \quad (9)$$

4.2 Pareto-Efficient Optimization

Transform to Multi-objective Optimization. Intuitively, we can regard w_k in Eq. (9) as hyper-parameters and find proper weights via grid search. However, the grid search operation is time-consuming especially when K is large, and even worse, there is no theoretical guarantee to ensure that the finding weights can lead to an optimal solution of θ . In fact, through revisiting Eq. (9) from a new perspective, we can naturally transform the multi-teacher distillation to a multi-objective optimization problem, in which each term represents a fairness-related objective for the student model. Specifically, the terms $\mathcal{L}_k(\theta)$ actually emphasize the student GNN model's capability in achieving the k th kind of fairness.

However, the conventional weighted sum method has been shown to be insufficient in handling non-convex problems [3], which may result in obtaining a sub-optimal solution. Therefore, we adopt a weighted metric method, which has been shown to alleviate non-convex optimization problems to some extent¹, to reformulate Eq. (9). The difference between the two kinds of methods is shown by toy graphs in Figure 3 (a)-(b). We also provide theoretical analysis to prove that the optimization of our new objective function can lead to the minimizing of the upper bound of the original objective function in Eq. (9). Specifically, the weighted metric-based objective function is:

$$\begin{aligned} \min_{\theta} \quad & \left(\sum_{k=1}^K w_k \mathcal{L}_k(\theta)^p \right)^{\frac{1}{p}} \\ \text{s.t.} \quad & \sum_{k=1}^K w_k = 1, w_k > 0 \quad \forall k \in \{1, 2, \dots, K\}, \end{aligned} \quad (10)$$

where $p \in \mathbb{R}^+$ is a constant, and Eq. (10) is equivalent to Eq. (9) when $p = 1$. Then we have:

LEMMA 2. *Minimizing the weighted metric-based objective leads to the minimization of the upper bound of the conventional weighted sum-based objective:*

$$\left(\sum_{k=1}^K w_k \mathcal{L}_k(\theta)^p \right)^{\frac{1}{p}} \geq \sum_{k=1}^K w_k \mathcal{L}_k(\theta). \quad (11)$$

¹Sudhof Scott, Lecture 9: Multi-Objective Optimization, <https://engineering.purdue.edu/~sudhoff/ee630/Lecture09.pdf>

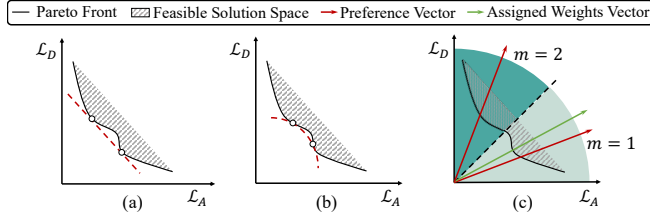


Figure 3: Toy graphs for illustrating the difference between (a) weighted sum method and (b) weighted metric method when facing non-convex solution space. (c) A toy graph for illustrating the concept of preference vectors where $M = 2$.

PROOF. The proof can be derived from the following inequalities:

$$\left(\sum_{k=1}^K w_k L_k(\theta)^p\right)^{\frac{1}{p}} \geq \left(\sum_{k=1}^K w_k L_k(\theta)\right)^{\frac{1}{p}} = \sum_{k=1}^K w_k L_k(\theta). \quad (12)$$

This follows from Jensen's inequality and the characteristic of x^p as an increasing convex function when $x \geq 0, p > 0$. $\square$

Pareto Efficient Conditions. Note that simply optimizing Eq. (10) without proper weights assigned cannot lead to a Pareto-efficient solution. Therefore, we further derive conditions on the scalarization weights that guarantee the Pareto efficiency of solutions. To be more precise, considering the difficulty of derivation when p is large, here we set p to 2. With the KKT conditions (Karush-Kuhn-Tucher Conditions) [3, 17], we can easily derive following equations from Eq. (10), with more details provided in the Appendix A.3:

$$\sum_{k=1}^K w_k = 1, \text{ and } \sum_{k=1}^K w_k L_k(\theta) \nabla_{\theta} L_k(\theta) = 0, \quad (13)$$

where $\nabla_{\theta} L_k(\theta)$ denotes the gradient of $L_k(\theta)$. The condition is equivalent to solving the following optimization problem:

$$\begin{aligned} \min \quad & \left\| \sum_{k=1}^K w_k L_k(\theta) \nabla_{\theta} L_k(\theta) \right\|_2^2 \\ \text{s.t.} \quad & \sum_{k=1}^K w_k = 1, \quad w_k > 0, \quad \forall k \in \{1, 2, \dots, K\}. \end{aligned} \quad (14)$$

By optimizing the above quadratic polynomial with fixed model parameters θ , proper weights could be assigned, which allow us to find solutions along the gradient direction of minimizing all objective functions simultaneously. Ideally, when the KKT conditions are satisfied, Pareto-efficient solutions are obtained [28]. Feasible solutions to this Quadratic Programming problem can be easily found through fast heuristics.³

4.3 Diverse Hybrid Fairness under Controls

Although we can get proper weights from optimizing Eq. (14) and then get near-Pareto-efficient solutions, the region of solutions is uncontrollable and lacks of diversity, especially after considering the large space of Pareto-efficient solutions. For example, without controls, a Pareto-efficient solution $\hat{\theta}$ may perform remarkably on minimizing the DEO on one specific fairness while having poor results in minimizing others. Some researchers [17] try to add some boundaries for each weight like $w_k \geq c_k, \forall k \in \{1, 2, \dots, K\}$,

³We use gurobi solver in this work. <https://www.gurobi.com/>

however, such manual operation is time-consuming and extremely relies on expertise. Similar to [19], we introduce the concept of *preference vector* to LibraGNN, which could control the student model to find diverse Pareto-efficient solutions in a coarse-grained manner. Specifically, here we show the setting of two objectives for better understanding, and the scenarios of more objectives will be discussed in the Appendix A.1. As shown in Figure 3(c), we first divide the whole solution space into M sub-regions uniformly ($M = 2$ in the figure), and the preference vector for the m th sub-region can be formulated as the corresponding sub-region's bisector, i.e.:

$$[\hat{w}_{1,m}, \hat{w}_{2,m}] = \left[\cos\left(\frac{(2m-1)\pi}{4M}\right), \sin\left(\frac{(2m-1)\pi}{4M}\right) \right], \quad (15)$$

where $[\hat{w}_{1,m}, \hat{w}_{2,m}]$ is the preference vector in the m th sub-region and $m \in \{1, 2, \dots, M\}$. To restrict the assigned weights derived from Eq. (14) in the m th sub-region, we further add a constraint on the cosine value of the angle between the assigned weights vector and the m th sub-region's preference vector to Eq. (14) as our final optimization objective:

$$\begin{aligned} \min \quad & \left\| \sum_{k=1}^2 w_k L_k(\theta) \nabla_{\theta} L_k(\theta) \right\|_2^2 \\ \text{s.t.} \quad & \sum_{k=1}^2 w_k = 1, \quad w_k > 0, \quad \forall k \in \{1, 2\} \\ & \sum_{k=1}^2 w_k * \hat{w}_{k,m} > \cos\left(\frac{\pi}{4M}\right) \left(\sum_{k=1}^2 w_k^2\right)^{\frac{1}{2}} \left(\sum_{k=1}^2 \hat{w}_{k,m}^2\right)^{\frac{1}{2}}. \end{aligned} \quad (16)$$

For a better understanding, we briefly summarize the whole training process of LibraGNN in Algorithm 1.

Algorithm 1 Training process for LibraGNN

Input: Graph $\mathcal{G}$ with labels. Hyper-parameters: number of sub-regions M and the preferred sub-region index m .

Output: An all-round GNN model with parameters θ

- 1: Train K different teacher models which have expertise in specific scenarios, and obtain their output logits $\mathbf{P}_k^T, \forall k \in \{1, 2, \dots, K\}$.
 - 2: Randomly initialize student GNN model with θ .
 - 3: Compute the preference vector through Eq. (15)
 - 4: Initialize the weights in Eq. (10) uniformly.
 - 5: **while** not converged **do**
 - 6: Compute $\mathcal{L}_k$ by Eq. (7).
 - 7: Update the parameters of student GNN model θ through minimizing Eq. (10).
 - 8: Compute new weights w_k by minimizing Eq. (16).
 - 9: Assign new weights w_k to Eq. (10).
 - 10: **end while**
 - 11: **return** θ
-

5 Experiment

5.1 Experimental Setup

Datasets. We follow the prior work of Dai et al. [7], and use two social graphs named **Pokec-z** and **Pokec-n**. Both datasets are sampled from Pokec dataset², the most popular social network in Slovakia, according to the provinces that users belong to. The datasets

²<https://snap.stanford.edu/data/soc-Pokec.html>

Table 1: Dataset statistics

Dataset	Users	Interactions	Attributes	Classes
Pokec-z	67,797	882,765	59	5 (working field)
Pokec-n	66,569	729,129	59	4 (working field)
Weibo	100,000	2,531,524	6	2 (verified type)

provide anonymized user profiles including ages, gender, and other attributes, and we treat gender as a sensitive attribute. The final classification task is to predict the working field of users. We also use a social network dataset **Weibo** [41], which is one of the most popular online social platforms in China. The original network contains 1.7 million users, and for comparisons with baselines that require high memory computation, we randomly sample a subgraph with 0.1 million users. The original dataset files can be downloaded here³. Each node in the dataset represents a user and each directed link represents a following relationship. Anonymized user profiles including gender and locations are also provided in the dataset, and we treat gender as the default sensitive attribute. We binarize the users into two categories according to their verified type for classification. Table 1 reports the detailed dataset statistics.

Evaluation. We propose several metrics to evaluate our models. For model performance, we report the node classification accuracy on the test set. For fairness, we report the DEO and the corresponding HFI to evaluate fairness and hybrid fairness, respectively. In the experiment, we mainly consider two types of group fairness in GNNs, namely *attribute-based fairness* and *degree-based fairness*. Another form of fairness like counterfactual fairness will also be included in the experiments. For attribute-based fairness, we divide nodes into two groups based on the sensitive attribute, gender, while for degree-based fairness, we group nodes according to whether their degrees exceed a specific threshold *deg*.

Base GNNs. As a model-agnostic framework, LibraGNN can work with different GNN backbones. In detail, we employ GCN [16] as the default GNN backbone, and two alternative base GNNs, namely SGC [36] and GraphSAGE [12] are also included in the experiments.

Baselines. We comprehensively consider three kinds of baselines, including (1) *Degree-based fair GNN*: TailGNN [21] and DegFairGNN [22]. (2) *Attribute-based fair GNN*: FairGNN [7], FairVGNN [35], EqGNN [29], and FairSIN [37]. (3) *Multi-objective algorithms*: Cascade, Ensemble, PE-LTR [17], CA-MKD [40], InterD [8], MTARD [42], LTE4G [39], CFC [2], and GradNorm [5].

Reproducibility. For Pokec-z and Pokec-n, we follow the settings of prior work [7] that only limited label information is available, and the training set size is fixed to 1000. For Weibo, we randomly select 20000 nodes and split them into training, validation, and test sets with a ratio of 2:1:1. Similar to the setup in TailGNN [21], the *deg* for Pokec-z and Pokec-n is set to be 5, and 20 for Weibo after considering its high density. For all student models, we set the hidden dimension to 128 and the layer number to 2. Dropout is set to 0. Learning rates are empirically set in a range from 0.002 to 0.01. We set M in {8, 10} and tune the sub-region index for the best hybrid fairness. The weight in knowledge distillation loss α is chosen from {0, 0.001, 0.01}.

³<https://www.aminer.cn/influencelocality>

5.2 Performance Comparison

In this section, we evaluate LibraGNN in comparison with baselines in terms of both accuracy and fairness. Specifically, with GCN [16] as the student model and backbone for other baselines, we choose DegFairGNN [22] and EqGNN [29], which have previously achieved state-of-the-art performance in degree-based fairness and attribute-based fairness, as teacher models. To distinguish the DEO results on degree-based fairness and attribute-based fairness, we abbreviate the metrics as DEO_d and DEO_a , respectively.

We report the average results with standard deviations in Table 2 after repeating each model five times, and we can have the following observations: (1) For degree-based fair GNNs and attribute-based fair GNNs, they tend to improve the fairness in a specific perspective while maintaining unfairness from the other view. For example, TailGNN effectively reduces the DEO_d on Pokec-z from 6.52% to 3.97% but slightly increases the DEO_a from 11.94% to 13.22%. This kind of observation supports our motivation to pursue hybrid fairness. (2) On three datasets, our proposed LibraGNN consistently achieves the highest HFI. Instead of heuristic optimization, we believe that the Pareto-guided optimization provides LibraGNN with a solid theoretical guarantee for finding near-optimal solutions, which can be supported by the best results of LibraGNN on HFI. (3) In some cases, LibraGNN could even achieve the best performance in a single fairness aspect. For example, the DEO_d of LibraGNN on Pokec-n is 0.94%, which is even better than that of its teacher model DegFairGNN. In fact, similar observations have been found in prior research on knowledge distillation [15]. Within the same backbone structure, the multi-teacher approach actually serves as a regularization mechanism during the training process, helping the student learn from a broader range of information than what is provided for teachers. (4) LibraGNN successfully inherits the merits of teacher models (i.e., DegFairGNN and EqGNN) in preserving utility while improving the fairness, thus improving hybrid fairness with only a minimal amount of utility compromise (less than 3%).

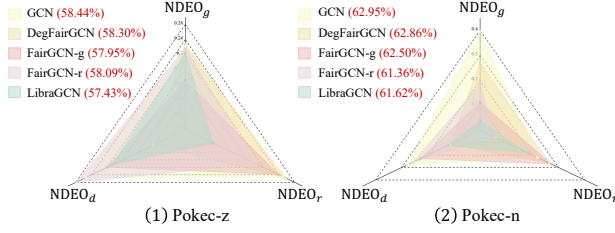
5.3 Analysis on Expandability

As a multi-teacher knowledge distillation framework, LibraGNN possesses the theoretical potential to handle various forms of fairness. To substantiate this claim, we further investigate the expandability of LibraGNN from the following two perspectives.

Expandability to the Number of Fairness. In addition to our existing experimental setup, we take the group fairness belonging to different regions into consideration. Specifically, we employ three teacher models, namely FairGCN-g, FairGCN-r, and DegFairGCN, to guide the student model (i.e. GCN) in mitigating unfairness based on gender, region, and node degree, respectively. Correspondingly, we designate the normalized DEO associated with degree, gender, and region as $NDEO_d$, $NDEO_g$, and $NDEO_r$, respectively. Note that, although FairGCN-g and FairGCN-r share the same model architecture, they use independent sets of parameters to handle different kinds of fairness. The accuracy of each model is reported alongside the legends. Figure 4 depicts the experimental results. It can be seen that LibraGNN consistently exhibits the top-2 results among the three types of unfairness, thus covering a considerably smaller area. This outcome confirms LibraGNN’s ability to address multiple unfairness concurrently and achieve hybrid fairness.

Table 2: Comparison with baselines in percentage (%). The best results on fairness are bold, and the runner-up is underlined.

	Pokec-z				Pokec-n				Weibo			
	Acc.↑	DEO _a ↓	DEO _d ↓	HFI↑	Acc.↑	DEO _a ↓	DEO _d ↓	HFI↑	Acc.↑	DEO _a ↓	DEO _d ↓	HFI↑
GCN	58.97±0.32	11.94±0.62	6.52±0.32	0.00±0.00	63.16±0.35	4.77±0.44	5.08±0.50	0.00±0.00	72.99±0.12	26.50±0.25	2.73±0.25	0.00±0.00
TailGNN	58.13±0.09	13.22±0.43	3.97±0.49	14.20±0.52	63.63±0.24	3.18±0.41	2.37±0.87	43.34±0.59	68.68±0.36	29.39±0.04	1.77±0.08	12.13±0.11
DegFairGNN	57.00±0.48	14.29±1.29	3.46±0.29	27.25±0.51	63.27±0.35	3.27±0.41	2.80±0.50	38.16±0.43	69.31±0.21	27.51±0.62	1.47±0.66	21.17±0.42
FairGNN	57.95±0.92	9.37±0.56	4.58±0.82	25.64±0.83	61.59±0.89	2.27±0.69	5.66±1.08	20.50±0.78	72.80±0.27	23.75±0.38	2.78±0.22	4.27±0.25
FairVGNN	58.05±0.36	7.44±0.41	5.18±0.52	29.12±0.62	62.48±0.33	<u>1.48</u> ±0.28	3.98±0.42	45.31±0.38	71.27±0.14	15.73±0.51	2.14±0.85	31.13±0.42
EqGNN	57.75±0.47	7.25±0.39	6.37±0.53	20.79±0.45	62.81±0.29	1.83±0.33	5.20±0.89	29.64±0.64	71.45±0.96	11.60±0.32	1.47±0.12	51.19±0.73
FairSIN	57.60±0.34	<u>6.73</u> ±0.31	5.36±0.42	30.71±0.31	62.45±0.21	1.83±0.29	2.05±0.18	60.64±0.58	69.08±0.69	22.06±0.43	0.89±0.08	42.08±0.52
Cascade Ensemble	58.12±0.45	12.43±0.41	<u>3.21</u> ±0.28	23.33±0.37	61.93±0.55	1.45 ±0.22	3.65±0.45	48.88±0.37	70.37±0.29	26.43±0.37	1.92±0.18	14.97±0.35
PE-LTR	58.34±0.37	14.82±1.02	5.28±0.78	−2.55±0.82	62.90±0.48	3.54±0.24	4.32±0.21	20.37±0.32	69.83±0.91	27.32±0.42	1.01±0.15	29.95±0.37
InterD	58.42±0.22	11.54±1.80	5.12±0.74	12.41±0.83	63.02±0.27	1.50±0.51	2.56±0.29	<u>59.08</u> ±0.46	70.94±0.23	14.08±1.01	<u>1.02</u> ±0.11	54.75±0.92
CA-MKD	58.58±0.41	12.00±0.85	5.29±0.11	9.18±0.35	62.80±0.15	2.95±0.43	2.32±0.22	46.24±0.55	71.25±0.52	<u>10.12</u> ±1.38	1.15±0.15	<u>59.84</u> ±0.87
MTARD	58.55±0.22	12.22±0.87	4.12±0.65	17.23±0.57	62.60±0.25	2.34±0.51	2.04±1.23	55.39±0.72	70.17±0.14	9.75 ±0.85	1.23±0.46	59.08±0.94
LTE4G	58.45±0.20	11.05±0.35	4.68±0.79	17.84±0.66	62.56±0.06	3.54±0.20	3.02±0.46	33.17±0.32	70.39±0.08	23.75±1.24	1.37±0.20	30.10±0.93
CFC	58.65±0.11	11.22±0.32	5.67±0.53	9.53±0.41	62.03±0.08	2.32±0.26	3.01±0.44	46.06±0.43	72.38±0.67	26.33±1.02	1.57±0.26	21.57±0.76
GradNorm	57.67±0.45	8.27±0.24	3.98±0.45	<u>34.85</u> ±0.39	62.14±0.22	3.27±0.30	<u>1.97</u> ±0.19	46.33±0.25	71.03±0.20	24.68±0.38	1.77±0.21	21.02±0.39
LibraGNN	57.52±0.62	11.25±0.23	2.53 ±0.35	33.49±0.29	62.81±0.29	1.77±0.34	3.97±0.27	42.37±0.30	70.95±0.53	23.56±0.41	1.83±0.33	22.01±0.38
Improv.	−2.95%	+73.70%	+36.50%	—	−0.47%	+47.59%	+81.50%	—	−2.41%	+53.74%	+84.62%	—

**Figure 4: Radar maps for measuring the expandability of LibraGNN. The accuracy of each model is reported with red fonts. Smaller triangles are better.**

Expandability to the Form of Fairness. One merit of LibraGNN is the capability to handle both group fairness and other forms of fairness, such as counterfactual fairness [13]. Specifically, we conduct experiments using NIFTY [1] as the teacher model for counterfactual fairness. The HFI is redefined by both DEO and DCP, whose definitions are given in Section 3. From Table 3 we can observe that, LibraGNN consistently achieves the top-2 performance on both DEO and DCP, which leads to the best results on HFI. This observation confirms the effectiveness of our approach in concurrently achieving different types of fairness.

5.4 Sensitivity Studies

We conduct comprehensive sensitivity studies in this section to explore whether LibraGNN is robust to a wide range of settings. In detail, we investigate LibraGNN’s sensitivity to the choice of student models, teacher models, and the ways of knowledge distillation.

Different Student Models. We first investigate the performance of LibraGNN on two alternative student GNN models, namely SGC [36] and GraphSAGE [12]. Specifically, we choose DegFairGNN [22] and FairGNN [7] as teacher models after considering their compatibility with different base GNNs. For a fair comparison, we

Table 3: Expandability to counterfactual fairness

	Pokec-z				Pokec-n			
	Acc.↑	DEO↓	DCP↓	HFI↑	Acc.↑	DEO↓	DCP↓	HFI↑
GCN	58.97	11.94	3.90	0	62.95	4.78	2.05	0
FairGCN	58.57	11.84	4.15	2.79	61.59	2.27	1.98	<u>27.96</u>
NIFTY	59.41	<u>11.58</u>	<u>3.27</u>	<u>9.58</u>	63.59	4.44	<u>1.68</u>	12.58
LibraGNN	60.04	11.51	2.69	17.31	64.00	<u>3.65</u>	0.45	50.84

Table 4: Results of distilling on different base GNNs in percentage (%). The best fairness-related results are bold, and the runner-up is underlined.

	Pokec-z				Pokec-n			
	Acc.↑	DEO _a ↓	DEO _d ↓	HFI↑	Acc.↑	DEO _a ↓	DEO _d ↓	HFI↑
SGC	58.44	13.06	6.13	0	62.95	4.78	5.28	0
DegFairSGC	58.40	11.42	3.66	26.43	62.91	3.20	<u>1.80</u>	<u>49.48</u>
FairSGC	58.67	<u>7.75</u>	<u>5.05</u>	<u>29.14</u>	63.68	<u>3.14</u>	3.55	33.54
LibraSGC	56.37	6.25	<u>5.05</u>	34.88	62.23	2.88	1.74	53.40
GraphSAGE	56.85	12.97	5.73	0	60.33	2.54	2.82	0
DegFairSAGE	55.67	13.42	1.64	33.95	60.36	2.06	0.24	<u>55.19</u>
FairSAGE	54.85	0.23	5.48	<u>51.29</u>	60.36	<u>1.14</u>	<u>2.25</u>	37.67
LibraSAGE	54.15	<u>0.70</u>	<u>4.33</u>	59.52	58.32	0.01	2.28	59.38

set the base GNN of teacher models to be identical to that of student models, and the results are presented in Table 4. It can be observed that, even after changing the student GNN model, LibraGNN still achieves the best performance in hybrid fairness (HFI), with only a minor compromise in utility. This finding confirms the compatibility of LibraGNN with different student GNN models.

Different Teacher Models. Besides DegFairGNN and EqGNN, we choose TailGNN and FairGNN as another pair of teacher models for comparison, and the denotations of teacher models are abbreviated

Table 5: Comparison of different teacher models and different distillation methods. The best results on fairness are bold. * denotes the abbreviation of the combination of teacher models, and † denotes the proposed settings in LibraGNN.

	Pokec-z				Pokec-n			
	Acc.↑	DEO _a ↓	DEO _d ↓	HFI↑	Acc.↑	DEO _a ↓	DEO _d ↓	HFI↑
GCN	58.97	11.94	6.52	0	63.16	4.77	5.08	0
Results of different teacher models								
T&F*	58.20	9.24	3.17	37.00	62.68	2.60	0.87	64.18
D&E*†	57.23	3.14	4.14	55.10	62.86	2.50	0.94	64.54
Results of different distillation methods								
MMD Loss	58.13	10.12	4.07	26.41	62.58	2.71	0.68	64.90
KL Loss	58.27	7.89	4.22	34.60	62.81	1.69	1.34	69.10
L2 Norm†	57.23	3.14	4.14	55.10	62.86	2.50	0.94	64.54

as D&E, T&F for conciseness. The student model is the GCN. As shown in Table 5, although the performance on reducing one single kind of fairness is different (i.e. DEO_a or DEO_d), the LibraGNN variant achieves comparable results on both HFI and accuracy, which indicates LibraGNN has strong robustness in finding Pareto-efficient solutions with different choices on teacher models.

Different Distillation Methods. In this part, we aim to explore the impact of different distillation methods on LibraGNN. To be specific, we analyze three kinds of distillation approaches, including L2-norm loss [26], KL loss [40, 42], and MMD loss [15]. As reported in Table 5, L2-norm loss and KL loss consistently achieve the best performance on hybrid fairness (HFI) while maintaining competitive accuracy compared with MMD loss. One possible reason for this is the difficulty in finding an appropriate kernel function that can accurately measure the maximum mean difference between distributions. Empirically, L2-norm loss or KL loss are preferable alternatives, and we choose L2-norm loss in LibraGNN because it can facilitate the proofs of some key lemmas.

5.5 Ablation Studies

We further conduct ablation studies to verify the effectiveness of the design of LibraGNN. Specifically, we consider the following three variants of LibraGNN:

Different Weighted Objective Functions. In LibraGNN, we adopt the weighted metric method instead of the simple weighted sum method in the design of the objective functions. To this end, we validate the effectiveness of this design on three datasets by replacing the weighted metric method with the weighted sum method, denoted as "LibraGNN-W". As shown in Figure 5, LibraGNN consistently outperforms the variant in terms of HFI, while maintaining competitive accuracy across three datasets. We believe that the main reason is that the weighted metric method can explore the Pareto efficiency in the non-convex solution space, whereas the weighted sum method is only applicable to convex problems.

Objective Simplification. We claim that given teacher models with high utility, there is no need to add cross-entropy loss functions in the objective functions of LibraGNN for maintaining high accuracy. In this part, we add the cross-entropy loss functions in the objective functions and denote it as "LibraGNN-S". From Figure 5 we can observe that, in most cases, the variants and full LibraGNN

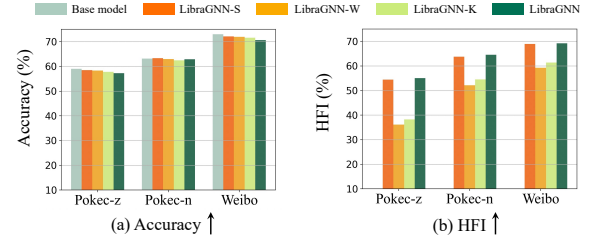


Figure 5: Ablation study of LibraGNN

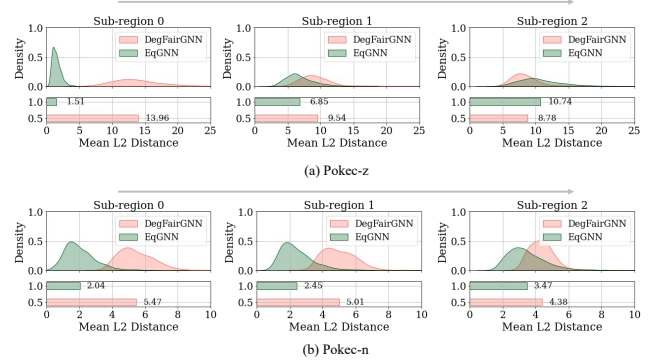


Figure 6: The visualization of the L2 distance distributions between the student model and two teacher models illustrates the preference shift of LibraGNN.

demonstrate competitive performance in terms of both accuracy and HFI. This observation supports our claims that there is no need to add utility-oriented loss functions in LibraGNN, especially after considering the increasing complexity of optimization when the number of objective functions increases.

Without Knowledge Distillation. With the knowledge distillation module, LibraGNN can draw fair knowledge from multiple fairness-aware GNNs simultaneously in a unified manner. In this part, we replace the distillation loss $\mathcal{L}_k$ in Eq. (8) with normal fairness constraints, denoted as "LibraGNN-K", and use Pareto efficient conditions for optimization guidance. In detail, we employ the fairness loss in [22] as the fairness constraint. As shown in Figure 5, the variant consistently falls behind full LibraGNN in terms of HFI. Considering the diversity of fairness and the different causes of unfairness, we believe that knowledge distillation is a more effective strategy to handle multiple fairness in a unified manner.

5.6 In-depth Model Analysis

In this section, we carry out a series of in-depth analyses for LibraGNN from the following perspectives:

Analysis on the Controllability. To validate the effectiveness of LibraGNN in controlling the student model's preference towards different teacher models, as introduced in Section 4.3, we plot the L2 distance between the student model's outputs and the two teacher model's outputs on Pokec-z and Pokec-n as density maps, as illustrated in Figure 6. Specifically, we set M in Eq. (15) to 3 and train three student models with different preferences by increasing the sub-region index m from 1 to 3 (i.e. shifting preference from teacher EqGNN to DegFairGNN). To further quantify the variation

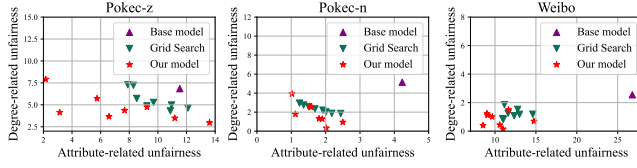


Figure 7: Comparison with different static weights

in the student model’s preference, we also display the mean L2 distance between the student model’s output embeddings and the two teacher models’ output embeddings under each density map. It can be seen that, as the sub-region index m increases, the output embeddings’ distance between the student model and EqGNN consistently grows larger, while the distance between the student model and DegFairGNN decreases, which means the student model gradually becomes more similar to DegFairGNN rather than EqGNN. These observations confirm that our LibraGNN can effectively control the student model’s preference towards different teachers.

Pareto Dominance Analysis. As a Pareto-guided framework, one expected merit of LibraGNN is that it can theoretically lead the model to find a Pareto-efficient solution. To verify our hypothesis, we compare the performance of LibraGNN with that through grid search. Specifically, we set the sub-region number M as 8, and tune m in Eq. (15) from 1 to 8. The results with the highest HFI under each setting are reported. To be fair, we also uniformly choose 8 different static weight pairs for comparison. Note that, all results reported are associated with a competitive accuracy, i.e. within a 3% relative gap compared with that of the backbone model. The results are shown in Figure 7, and we can find two interesting observations: (1) LibraGNN not only provides diverse solutions but also visually demonstrates the formation of the Pareto frontier, which dominates the results of grid search in most of the cases. (2) In some cases, the differences between LibraGNN and grid search are extremely small, because both two methods deploy scalarization methods, and theoretically have the same upper bound.

Case Studies. For further analysis of the effectiveness of LibraGNN, we conduct case studies on two specific users on Pokec-z, and the results are illustrated in Figure 8. The user statistics are listed on the left, in which the *Interaction* denotes the degree of a user, and *Working field* denotes the category to be predicted. To better understand the meanings of user profiles, we add two arrows besides to show if the user belongs to the advantaged group or disadvantaged group ($\uparrow$ denotes the advantaged group and vice versa). From Figure 8 we can observe that:

- For user #482, although it belongs to males, the disadvantaged group, with its high degree and abundant neighborhood information, EqGNN could successfully give a correct prediction after addressing the discrimination based on gender. However, since DegFairGNN is mainly designed to address the unfairness based on degree, it fails to overcome the inferiority of the disadvantaged group (males) and gives a wrong prediction for user #482. Fortunately, LibraGNN successfully learns the correct information from EqGNN, which can be verified by the similarity between the prediction distributions of EqGNN and LibraGNN, and gives a correct prediction.



Figure 8: Case studies on Pokec-z. Here we first provide three users’ statistics on the left, and then illustrate the corresponding prediction distributions from two teacher models and LibraGNN on the right. The ✓ and ✗ indicate whether the prediction results are correct.

- For user #173, it belongs to Females, the advantaged group, but it lacks neighborhood information because of its low degree. As a result, EqGNN makes the wrong prediction and is torn between category #1 and category #4. Such confusion is addressed in DegFairGNN due to its strong capability in achieving degree-related fairness. It also leads to the correct prediction of LibraGNN by aggregating the knowledge from both two teachers.
- We also find that in some cases LibraGNN are able to provide correct prediction for users belonging to multiple disadvantaged groups simultaneously. Take user #14605 as an example, since it belongs to males with low degrees, both two teacher models fail to predict correctly. However, with the merit of multi-teacher knowledge distillation, after aggregating the prediction distribution from both teacher models, LibraGNN successfully figures out the true label.

Other Analysis. Due to space limitations, we provide more in-depth model analyses in Appendix A.5.

6 Conclusion

In this paper, we focus on the issue of hybrid fairness in GNNs. In contrast to existing works that focus on fairness in a single aspect, our proposed LibraGNN could achieve a Pareto-efficient solution among different kinds of fairness. To better control the GNN’s preference for different fairness aspects, we introduce a preference vector module to constrain the optimization direction and find diverse solutions. Extensive experiments on multiple kinds of fairness consistently demonstrate that LibraGNN can significantly improve the model’s hybrid fairness with trivial utility compromise, while retaining good expandability and robustness.

Acknowledgments

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References

- [1] C. Agarwal, H. Lakkaraju, and M. Zitnik. Towards a unified framework for fair and stable graph representation learning. In *Proceedings of the 37th Conference on Uncertainty in Artificial Intelligence, UAI 2021, Virtual Event, July 27–30, 2021*, pages 2114–2124. AUAI Press, 2021.
- [2] A. J. Bose and W. L. Hamilton. Compositional fairness constraints for graph embeddings. In *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, Long Beach, California, USA, June 9–15, 2019*, pages 715–724. PMLR, 2019.
- [3] S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.
- [4] K. Burkholder, K. Kwock, Y. Xu, J. Liu, C. Chen, and S. Xie. Certification and trade-off of multiple fairness criteria in graph-based spam detection. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management, CIKM 2021, Virtual Event, Queensland, Australia, November 1–5, 2021*, pages 130–139. ACM, 2021.
- [5] Z. Chen, V. Badrinarayanan, C. Lee, and A. Rabinovich. GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *Proceedings of the 35th International Conference on Machine Learning, ICML 2018, Stockholmsmässan, Stockholm, Sweden, July 10–15, 2018*, pages 793–802. PMLR, 2018.
- [6] Z. Chen, T. Xiao, and K. Kuang. BA-GNN: On learning bias-aware graph neural network. In *Proceedings of the 38th IEEE International Conference on Data Engineering, ICDE 2022, Kuala Lumpur, Malaysia, May 9–12, 2022*, pages 3012–3024. IEEE, 2022.
- [7] E. Dai and S. Wang. Say no to the discrimination: Learning fair graph neural networks with limited sensitive attribute information. In *Proceedings of the 14th ACM International Conference on Web Search and Data Mining, WSDM 2021, Virtual Event, Israel, March 8–12, 2021*, pages 680–688. ACM, 2021.
- [8] S. Ding, F. Feng, X. He, J. Jin, W. Wang, Y. Liao, and Y. Zhang. Interpolative distillation for unifying biased and debiased recommendation. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2022, Madrid, Spain, July 11–15, 2022*, pages 40–49. ACM, 2022.
- [9] Y. Dong, J. Kang, H. Tong, and J. Li. Individual fairness for graph neural networks: A ranking based approach. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, KDD 2021, Virtual Event, Singapore, August 14–18, 2021*, pages 300–310. ACM, 2021.
- [10] Y. Dong, N. Liu, B. Jalaian, and J. Li. EDITS: Modeling and mitigating data bias for graph neural networks. In *Proceedings of the 31st ACM Web Conference, WWW 2022, Virtual Event, Lyon, France, April 25–29, 2022*, page 1259–1269. ACM, 2022.
- [11] Y. Dong, B. Zhang, Y. Yuan, N. Zou, Q. Wang, and J. Li. RELIANT: Fair knowledge distillation for graph neural networks. In *Proceedings of the 23rd SIAM International Conference on Data Mining, SDM 2023, Minneapolis-St. Paul Twin Cities, MN, USA, April 27–29, 2023*, pages 154–162. SIAM, 2023.
- [12] W. L. Hamilton, Z. Ying, and J. Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems, NeurIPS 2017, Long Beach, CA, USA, December 4–9, 2017*, pages 1024–1034. MIT Press, 2017.
- [13] S. Han, S. Lee, F. Wu, S. Kim, C. Wu, X. Wang, X. Xie, and M. Cha. DualFair: Fair representation learning at both group and individual levels via contrastive self-supervision. In *Proceedings of the 32nd ACM Web Conference 2023, WWW 2023, Austin, TX, USA, April 30–May 4, 2023*, pages 3766–3774. ACM, 2023.
- [14] G. E. Hinton, O. Vinyals, and J. Dean. Distilling the knowledge in a neural network. *CoRR*, abs/1503.02531, 2015.
- [15] S. Jung, D. Lee, T. Park, and T. Moon. Fair feature distillation for visual recognition. In *Proceedings of the 2021 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2021, virtual, June 19–25, 2021*, pages 12115–12124. Computer Vision Foundation / IEEE, 2021.
- [16] T. N. Kipf and M. Welling. Semi-supervised classification with graph convolutional networks. In *Proceedings of the 5th International Conference on Learning Representations, ICLR 2017, Toulon, France, April 24–26, 2017*. openreview.net, 2017.
- [17] X. Lin, H. Chen, C. Pei, F. Sun, X. Xiao, H. Sun, Y. Zhang, W. Ou, and P. Jiang. A pareto-efficient algorithm for multiple objective optimization in e-commerce recommendation. In *Proceedings of the 13th ACM Conference on Recommender Systems, RecSys 2019, Copenhagen, Denmark, September 16–20, 2019*, pages 20–28. ACM, 2019.
- [18] X. Lin, J. Kang, W. Cong, and H. Tong. BeMap: Balanced message passing for fair graph neural network. In *Proceedings of the Learning on Graphs Conference, LoG 2023, Virtual Event, November 27–30, 2023*. PMLR, 2023.
- [19] X. Lin, H. Zhen, Z. Li, Q. Zhang, and S. Kwong. Pareto multi-task learning. In *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, Vancouver, BC, Canada, December 8–14, 2019*, pages 12037–12047. MIT Press, 2019.
- [20] H. Ling, Z. Jiang, Y. Luo, S. Ji, and N. Zou. Learning fair graph representations via automated data augmentations. In *Proceedings of the 11th International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1–5, 2023*. openreview.net, 2023.
- [21] Z. Liu, T. Nguyen, and Y. Fang. Tail-GNN: Tail-node graph neural networks. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, KDD 2021, Virtual Event, Singapore, August 14–18, 2021*, pages 1109–1119. ACM, 2021.
- [22] Z. Liu, T. Nguyen, and Y. Fang. On generalized degree fairness in graph neural networks. In *Proceedings of the 37th AAAI Conference on Artificial Intelligence, AAAI 2023, Washington, DC, USA, February 7–14, 2023*, pages 4525–4533. AAAI Press, 2023.
- [23] Z. Luo, H. Huang, J. Lian, X. Song, X. Xie, and H. Jin. Cross-links matter for link prediction: Rethinking the debiased GNN from a data perspective. In *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10–16, 2023*. MIT Press, 2023.
- [24] Z. Luo, H. Huang, Y. Zhou, J. Zhang, N. Chen, and H. Jin. Are Your Models Still Fair? fairness attacks on graph neural networks via node injections. In *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10–15, 2024*. MIT Press, 2024.
- [25] Z. Luo, J. Lian, H. Huang, H. Jin, and X. Xie. Ada-GNN: Adapting to local patterns for improving graph neural networks. In *Proceedings of the 15th ACM International Conference on Web Search and Data Mining, WSDM 2022, Virtual Event / Tempe, AZ, USA, February 21–25, 2022*, pages 638–647. ACM, 2022.
- [26] A. Romero, N. Ballas, S. E. Kahou, A. Chassang, C. Gatta, and Y. Bengio. FitNets: Hints for thin deep nets. In *Proceedings of the 3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7–9, 2015*. openreview.net, 2015.
- [27] B. Rozemberczki, C. T. Hoyt, A. Gogoleva, P. Grabowski, K. Karis, A. Lamov, A. Nikolov, S. Nilsson, M. Ughetto, Y. Wang, T. Derr, and B. M. Gyor. ChemicalX: A deep learning library for drug pair scoring. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, KDD 2022, Washington, DC, USA, August 14–18, 2022*, pages 3819–3828. ACM, 2022.
- [28] O. Sener and V. Koltun. Multi-task learning as multi-objective optimization. In *Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems 2018, NeurIPS 2018, Montréal, Canada, December 3–8, 2018*, pages 525–536. MIT Press, 2018.
- [29] U. Singer and K. Radinsky. EqGNN: Equalized node opportunity in graphs. In *Proceedings of the 36th AAAI Conference on Artificial Intelligence, AAAI 2022, Virtual Event, February 22–March 1, 2022*, pages 8333–8341. AAAI Press, 2022.
- [30] W. Song, Y. Dong, N. Liu, and J. Li. GUIDE: Group equality informed individual fairness in graph neural networks. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, KDD 2022, Washington, DC, USA, August 14–18, 2022*, pages 1625–1634. ACM, 2022.
- [31] X. Song, J. Lian, H. Huang, Z. Luo, W. Zhou, X. Lin, M. Wu, C. Li, X. Xie, and H. Jin. xGNN: An extreme graph convolutional network for large-scale social link prediction. In *Proceedings of the 32nd ACM Web Conference 2023, WWW 2023, Austin, TX, USA, 30 April 2023 - 4 May 2023*, pages 349–359. ACM, 2023.
- [32] X. Song, J. Lian, H. Huang, M. Wu, H. Jin, and X. Xie. Friend recommendations with self-rescaling graph neural networks. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, KDD 2022, Washington, DC, USA, August 14–18, 2022*, pages 3909–3919. ACM, 2022.
- [33] S. Vandenhende, S. Georgoulis, and L. V. Gool. MTI-Net: Multi-scale task interaction networks for multi-task learning. In *Proceedings of the 16th European Conference on Computer Vision, ECCV 2020, Glasgow, UK, August 23–28, 2020*, pages 527–543. Springer, 2020.
- [34] N. Wang, L. Lin, J. Li, and H. Wang. Unbiased graph embedding with biased graph observations. In *Proceedings of the 31st ACM Web Conference 2022, WWW 2022, Virtual Event, Lyon, France, April 25–29, 2022*, pages 1423–1433. ACM, 2022.
- [35] Y. Wang, Y. Zhao, Y. Dong, H. Chen, J. Li, and T. Derr. Improving fairness in graph neural networks via mitigating sensitive attribute leakage. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, KDD 2022, Washington, USA, August 14–18, 2022*, pages 1938–1948. ACM, 2022.
- [36] F. Wu, A. H. S. Jr., T. Zhang, C. Fifty, T. Yu, and K. Q. Weinberger. Simplifying graph convolutional networks. In *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, Long Beach, California, USA, June 9–15, 2019*, pages 6861–6871. PMLR, 2019.
- [37] C. Yang, J. Liu, Y. Yan, and C. Shi. FairSIN: Achieving fairness in graph neural networks through sensitive information neutralization. In *Proceedings of the 38th AAAI Conference on Artificial Intelligence, AAAI 2024, Vancouver, Canada, February 20–27, 2024*, pages 9241–9249. AAAI Press, 2024.
- [38] Y. Yang, J. Qiu, M. Song, D. Tao, and X. Wang. Distilling knowledge from graph convolutional networks. In *Proceedings of the 2020 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2020, Seattle, WA, USA, June 13–19, 2020*, pages 7072–7081. Computer Vision Foundation / IEEE, 2020.
- [39] S. Yun, K. Kim, K. Yoon, and C. Park. LTE4G: Long-tail experts for graph neural networks. In *Proceedings of the 31st ACM International Conference on Information and Knowledge Management, CIKM 2022, Atlanta, GA, USA, October 17–21, 2022*, pages 2434–2443. ACM, 2022.

- [40] H. Zhang, D. Chen, and C. Wang. Confidence-aware multi-teacher knowledge distillation. In *Proceedings of the 2022 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2022, Virtual and Singapore, May 23–27, 2022*, pages 4498–4502. IEEE, 2022.
- [41] J. Zhang, B. Liu, J. Tang, T. Chen, and J. Li. Social influence locality for modeling retweeting behaviors. In *Proceedings of the 23rd International Joint Conference on Artificial Intelligence, IJCAI 2013, Beijing, China, August 3–9, 2013*, pages 2761–2767. IJCAI/AAAI, 2013.
- [42] S. Zhao, J. Yu, Z. Sun, B. Zhang, and X. Wei. Enhanced accuracy and robustness via multi-teacher adversarial distillation. In *Proceedings of the 17th European Conference on Computer Vision, ECCV 2022, Tel Aviv, Israel, October 23–27, 2022*, pages 585–602. Springer, 2022.
- [43] Y. Zhu, J. Li, L. Chen, and Z. Zheng. The devil is in the data: Learning fair graph neural networks via partial knowledge distillation. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining, WSDM 2024, Merida, Mexico, March 4–8, 2024*, pages 1012–1021. ACM, 2024.

A Appendix

A.1 Preference Vector for Multiple Fairness

In this part, we discuss the details of how to design a preference vector under the scenarios of multiple fairness.

Specifically, denote the preference vector as $\hat{\mathbf{w}} = [\hat{w}_1, \hat{w}_2, \dots, \hat{w}_K]$, and the assigned weights vector generated by LibraGNN as $\mathbf{w} = [w_1, w_2, \dots, w_K]$, where K denotes the number of fairness to be achieved. Considering the difficulty of directly finding evenly distributed sub-regions in the K -dimensional space, we turn to define the sub-regions and the corresponding preference vectors with $K-1$ two-dimensional weight pairs, i.e. $\langle \hat{w}_1, \hat{w}_2 \rangle, \langle \hat{w}_2, \hat{w}_3 \rangle, \dots, \langle \hat{w}_{K-1}, \hat{w}_K \rangle$. To be concrete, for k th pair of objectives, we can divide the two-dimensional space into M_k sub-regions uniformly, and then the m th sub-region's preference vector can be formulated as:

$$[\hat{w}_{k,m_k}, \hat{w}_{k+1,m_k}] = \left[\cos\left(\frac{(2m_k-1)\pi}{4M_k}\right), \sin\left(\frac{(2m_k-1)\pi}{4M_k}\right) \right], \quad (17)$$

where M_k denotes the number of sub-regions in the k th pair of objectives. Note that, for each pair of objectives, the preferred sub-region index m would also be different, so we denote the preferred sub-region index for the k th pair of objectives as m_k . By decomposing the K -dimensional space to $K-1$ two-dimensional space, we can evenly find sub-regions for each pair of objectives and set the corresponding preference vector. Through considering all $K-1$ preference vectors together, we can achieve diverse solutions in the K -dimensional space approximately. Similar to Eq.(11) in the main text, the objective functions under scenarios of K different fairness can be formulated as:

$$\begin{aligned} \min \quad & \left\| \sum_{k=1}^2 w_k L_k(\theta) \nabla_{\theta} L_k(\theta) \right\|_2^2 \\ \text{s.t.} \quad & \sum_{k=1}^K w_k = 1, \quad w_k > 0, \quad \forall k \in \{1, 2, \dots, K\} \\ & \sum_{k=1}^2 w_k * \hat{w}_{k,m_k} > \cos\left(\frac{\pi}{4M_1}\right) \left(\sum_{k=1}^2 w_k^2 \right)^{\frac{1}{2}} \left(\sum_{k=1}^2 \hat{w}_{k,m_k}^2 \right)^{\frac{1}{2}} \quad (18) \\ & \dots \\ & \sum_{k=1}^K w_k * \hat{w}_{k,m_k} > \cos\left(\frac{\pi}{4M_{K-1}}\right) \left(\sum_{k=1}^K w_k^2 \right)^{\frac{1}{2}} \left(\sum_{k=1}^K \hat{w}_{k,m_k}^2 \right)^{\frac{1}{2}} \end{aligned}$$

Since we need to select a sub-region and assign preference weights for each pair of objectives, there will be $\prod_{i=1}^{K-1} M_i$ different choices in total, which means when the K is large, compared with

grid search, it would be also very time-consuming to set a proper preference vector. This conclusion is in line with prior work [19].

A.2 Proof

LEMMA 3. *Given a teacher model with high utility, whose output logits are $\mathbf{P}_k^T \in \mathbb{R}^{N \times |\mathcal{Y}|}$, then minimizing objective $\mathcal{L}_k^1$ will lead to the minimization of $\mathcal{L}_{CE}$ between student model output $\mathbf{P}^S \in \mathbb{R}^{N \times |\mathcal{Y}|}$ and the one-hot true labels $\mathbf{P}^* \in \mathbb{R}^{N \times |\mathcal{Y}|}$.*

PROOF. For conciseness, we proceed with the proof from the instance level. Given the same sample, we denote the corresponding post-softmax output logits from the teacher model, and student model as $\mathbf{p}_k^T \in \mathbb{R}^{1 \times |\mathcal{Y}|}$ and $\mathbf{p}^S \in \mathbb{R}^{1 \times |\mathcal{Y}|}$, respectively. The corresponding one-hot true label is denoted as $\mathbf{p}^* \in \mathbb{R}^{1 \times |\mathcal{Y}|}$.

Let us first consider the following inequalities:

$$\begin{aligned} \|\mathbf{p}^S - \mathbf{p}^*\|_2 &= \left(\sum_{i=1}^{|\mathcal{Y}|} (p_i^S - p_i^*)^2 \right)^{\frac{1}{2}} \\ &= \left(\sum_{i=1}^{|\mathcal{Y}|} (p_i^S - p_{k,i}^T + p_{k,i}^T - p_i^*)^2 \right)^{\frac{1}{2}} \\ &\leq \left(\sum_{i=1}^{|\mathcal{Y}|} (p_i^S - p_{k,i}^T)^2 \right)^{\frac{1}{2}} + \left(\sum_{i=1}^{|\mathcal{Y}|} (p_{k,i}^T - p_i^*)^2 \right)^{\frac{1}{2}} \\ &= \|\mathbf{p}^S - \mathbf{p}_k^T\|_2 + \|\mathbf{p}_k^T - \mathbf{p}^*\|_2, \end{aligned} \quad (19)$$

where p_i denotes the logit on each category and the inequality can be easily derived from the Cauchy-Schwarz inequality after quadratic expansion. Considering the fact that $\|\mathbf{p}_k^T - \mathbf{p}^*\|_2$ has been fixed in advance to a constant, minimizing the distance between $\mathbf{p}^S$ and $\mathbf{p}_k^T$ can lead to a smaller upper bound of $\|\mathbf{p}^S - \mathbf{p}^*\|_2$.

We now aim to establish the reduction in the cross-entropy loss $\mathcal{L}_{CE} = H(\mathbf{p}^*, \mathbf{p}^S)$ as the L2 norm distance $\|\mathbf{p}^S - \mathbf{p}^*\|_2$ diminishes. The cross-entropy can be formulated as:

$$H(\mathbf{p}^*, \mathbf{p}^S) = - \sum_{i=1}^{|\mathcal{Y}|} p_i^* \log(p_i^S). \quad (20)$$

Since $\mathbf{p}_i^*$ is a one-hot vector, we can further simplify Eq (20) as:

$$H(\mathbf{p}^*, \mathbf{p}^S) = -p_y^* \log(p_y^S) \propto -p_y^S, \quad (21)$$

where y denotes the true label index. As $\|\mathbf{p}^S - \mathbf{p}^*\|_2$ minimize, the logit on the true label index p_y^S will increase to 1, while others approach 0, thus the cross-entropy loss $\mathcal{L}_{CE}$ decreases. $\square$

A.3 Derivation of KKT Conditions

Here we would like to further introduce the details for deriving Eq. (13). We first transform the original constrained optimization problem in Eq. (10) into an equivalent unconstrained optimization problem using the Lagrange multiplier method:

$$\min_{\theta, \lambda} F(\theta, \lambda),$$

where F is the Lagrangian function and is defined as:

$$F(\theta, \lambda) = \left(\sum_{k=1}^K w_k \mathcal{L}_k(\theta) \right)^{1/p} + \lambda \left(\sum_{k=1}^K w_k - 1 \right),$$

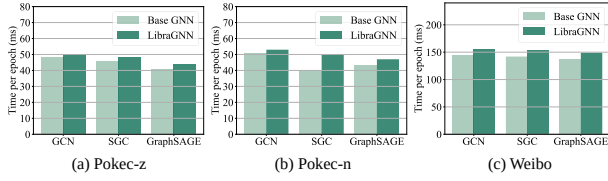


Figure I: The average training time per epoch of base GNN models and LibraGNN on three datasets

where λ denotes the Lagrange multiplier.

By taking the partial derivative of F for θ and setting it to zero, we can obtain the necessary condition for the optimal solutions:

$$\nabla_{\theta} F(\theta, \lambda) = \nabla_{\theta} \left(\sum_{k=1}^K w_k \mathcal{L}_k(\theta)^p \right)^{1/p} + \nabla_{\theta} \lambda \left(\sum_{k=1}^K w_k - 1 \right) \quad (22)$$

$$= \nabla_{\theta} \left(\sum_{k=1}^K w_k \mathcal{L}_k(\theta)^p \right)^{1/p} = 0. \quad (23)$$

As introduced in the paper, we set p as 2 and then we can derive the following equations through chain rules:

$$\left(\sum_{k=1}^K w_k \mathcal{L}_k(\theta) \right)^{-\frac{1}{2}} * \left(\sum_{k=1}^K w_k \mathcal{L}_k(\theta) \nabla_{\theta} \mathcal{L}_k(\theta) \right) = 0. \quad (24)$$

Since the first term can not be zero, now we can successfully obtain Eq. (13), i.e., $\sum_{k=1}^K w_k \mathcal{L}_k(\theta) \nabla_{\theta} \mathcal{L}_k(\theta) = 0$.

A.4 Discussion on Selecting Teacher Models

One key step of LibraGNN is to select proper fairness-aware teacher GNN models at first. In this section, we would like to provide two insightful suggestions for practitioners to choose appropriate teacher models for LibraGNN.

- **Selecting accurate teachers.** As we emphasized in Section 4.1, LibraGNN will not only inherit the fairness-related knowledge but also the utility of teacher models through knowledge distillation. Therefore, it is important to choose teacher models with high utility in advance, thus preserving the utility of LibraGNN. For the convenience of future deployment, we have provided several utility-friendly fair GNN models in Table 2, such as FairGNN [7], FairVGNN [35], FairSIN [37], and DegFairGNN [22], which hopefully sheds light on the teacher model selection.
- **Selecting efficient teachers.** In the context of multi-teacher knowledge distillation, we acknowledge that LibraGNN will inevitably incur additional time overhead during pre-training teacher models. In this way, it is also crucial to select efficient fair models as teachers to reduce potential time overhead. Fortunately, we empirically find several efficient fairness-aware GNN models in this work, such as FairGNN [7] and FairSIN [37] for attribute-related fairness, and DegFairGNN [22] for degree-related fairness. We will elaborate on this claim with a more detailed experimental analysis in the following parts.

In summary, when selecting teacher models for LibraGNN, it is best to choose models that combine high utility and efficiency. We have provided some model selection experiences and references in the above discussion, hoping to facilitate the practical use of LibraGNN in the future.

Table I: Training time cost per epoch of several well-established fair GNN models

	FairGNN	FairSIN	DegFairGNN
Pokec-z	0.0690s	0.6633s	0.1237s
Pokec-n	0.0617s	0.6049s	0.1192s
Weibo	0.2033s	1.6878s	0.2282s

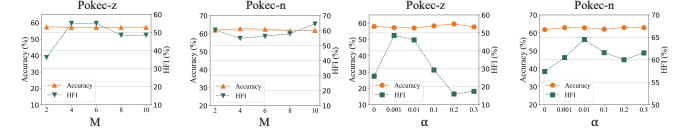


Figure II: Hyper-parameter analysis of the number of sub-regions M and weight α on datasets Pokec-z and Pokec-n

A.5 Additional In-depth Model Analysis

Efficiency Analysis. In this part, we conduct experiments to analyze the efficiency of LibraGNN. Specifically, we illustrate the training time (ms) per epoch of three base GNN models, including GCN, SGC, and GraphSAGE, and the corresponding LibraGNNs. As shown in Figure I, we can observe that our method will only increase a little time cost compared with the base GNNs. This matches our expectations because the step of computing weights that satisfy Pareto efficiency will only take a limited time, due to the high efficiency of fast heuristics.

As we discussed in Section A.4, we also conduct experiments to analyze the time complexity of some feasible teacher models for deploying LibraGNN in Table I. It can be seen that, all the listed teacher models exhibit high efficiency during the training process, which effectively avoids the potential heavy time overhead during teacher pre-training. Note that, here we only empirically provide several feasible teacher models with high efficiency for the convenience of deploying LibraGNN, and we also encourage practitioners to find more powerful and efficient teacher models in practice in the future.

Hyper-parameter Analysis. Here we would like to analyze the impact of several key hyper-parameters to LibraGNN, including the number of sub-region M and the weight α in the knowledge distillation objective function.

- **The number of sub-regions M .** We tune M in a range of $\{2, 4, 6, 8, 10\}$, and report the best HFI index with comparable utility. The student model is GCN. The impact of M on LibraGNN is shown in Figure II, and we can have the following observations: (1) The utility of LibraGNN is robust to M and can always maintain competitive accuracy. (2) Although different datasets show slightly different trends for the choice of M , we empirically find that $M \geq 6$ would lead to a near-optimal performance on both utility and hybrid fairness.
- **Weight α in knowledge distillation.** To analyze the impact of α , we tune it in $\{0, 0.001, 0.01, 0.1, 0.2, 0.3\}$ and report the accuracy and HFI index on Pokec-z and Pokec-n in Figure II. The student model is GCN and other hyper-parameters are set by default. It can be seen that, different datasets generally have different optimal settings about α . Empirically, $\alpha \in \{0.001, 0.01\}$ will lead to decent utility and hybrid fairness performance.